

Dataset Analysis Approach

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 sicor_municipal_co... .csv
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.



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To analyze the dataset and address your request, I will first load the dataset and examine its structure, key characteristics, and contents. Once I understand the dataset, I can proceed to answer specific analytical questions using Python code.

Let me start by loading the dataset and inspecting its structure.

Step 1: Load and Inspect the Dataset

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows of the Dataset:")
13 print(data.head())
```

Key Characteristics of the Dataset

From the output of the above code, we can identify:

1. **Number of Rows and Columns** : Total records and fields in the dataset.
2. **Column Names and Data Types** : What kind of data each column contains (e.g., numeric, categorical, dates).
3. **Missing Values** : Presence of null or missing values in any column.
4. **Unique Values** : Identifying categorical or identifier columns.
5. **Summary Statistics** : For numeric columns, understanding their range, mean, median, etc.

Once I have this information, I can proceed to address a specific analytical question. Please provide the analytical question you'd like answered using this dataset.

Is it possible to predict the top 10 financial institutions that will receive the most investment for the rest of the year based on data from the first semester? Define three approaches to address this issue and select the best one.



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To address the question of predicting the top 10 financial institutions that will receive the most investment for the rest of the year based on data from the first semester, we can consider three distinct approaches. Each approach has its strengths and limitations, and I will evaluate them to select the best one.

Approach 1: Historical Trend Analysis

Description:



This approach involves analyzing historical data from the first semester to identify trends in investment allocations. Specifically:

1. Aggregate the total investment received by each financial institution during the first semester.
2. Rank the institutions based on their total investment.
3. Assume that the trend observed in the first semester will continue into the rest of the year.
4. Use this ranking to predict the top 10 institutions likely to receive the most investment.

Implementation Steps:

1. Filter the dataset to include only records from the first semester.
2. Group the data by financial institution (e.g., using a unique identifier column).
3. Sum the total investment (`contract_value` or equivalent) for each institution.
4. Sort the institutions in descending order of total investment.
5. Select the top 10 institutions.

Strengths:

- Simple and straightforward.
- Does not require complex modeling or assumptions.
- Leverages actual data from the first semester.

Limitations:

- Assumes that past trends will persist, which may not account for sudden changes in market conditions, policies, or external factors.
- Ignores potential seasonality or anomalies in the first semester.

Approach 2: Machine Learning-Based Prediction

Description:

This approach uses machine learning to predict future investments based on patterns in the first semester's data. The steps include:

1. Preprocess the data to extract features such as:
 - Total investment per institution.
 - Number of contracts awarded.
 - Average contract size.
 - Sector or category of contracts.
 - Geographic region of the institution.
2. Train a regression or classification model to predict future investment amounts for each institution.
3. Use the trained model to forecast investments for the rest of the year.
4. Rank institutions based on predicted investment amounts and select the top 10.

Implementation Steps:

1. Preprocess the dataset to extract relevant features and target variables (e.g., total investment).
2. Split the data into training and testing sets.
3. Train a model (e.g., Random Forest, Gradient Boosting, or Linear Regression) using the first semester's data.
4. Evaluate the model's performance on the test set.
5. Use the trained model to predict future investments and rank institutions.

Strengths:

- Can capture complex relationships and interactions between variables.
- Allows for more accurate predictions if the model is well-trained and validated.
- Can incorporate external factors (e.g., economic indicators) if available.

Limitations:

- Requires sufficient data and computational resources.
- May overfit if the dataset is small or noisy.



may extend to the dataset is small or noisy.

- Relies on the assumption that the relationships in the first semester's data will hold for the rest of the year.

Approach 3: Expert Judgment with Data Support

Description:

This approach combines data analysis with expert judgment to predict the top 10 institutions. It involves:

1. Analyzing the first semester's data to identify key trends and patterns.
2. Consulting domain experts (e.g., economists, financial analysts) to assess factors that could influence future investments, such as:
 - Changes in government policies.
 - Economic forecasts.
 - Institutional reputation and capacity.
3. Combining expert insights with data-driven rankings to refine the prediction.

Implementation Steps:

1. Perform a preliminary analysis of the first semester's data to identify leading institutions.
2. Conduct interviews or surveys with experts to gather qualitative insights.
3. Adjust the rankings based on expert input, considering factors like policy shifts or emerging opportunities.
4. Finalize the list of top 10 institutions.

Strengths:

- Incorporates both quantitative and qualitative insights.
- Flexible and adaptable to changing circumstances.
- Accounts for external factors that data alone may not capture.

Limitations:

- Subjective and potentially biased due to reliance on expert opinions.
- Time-consuming and resource-intensive.
- Difficult to scale or automate.

Comparison of Approaches

CRITERIA	HISTORICAL TREND ANALYSIS	MACHINE LEARNING-BASED PREDICTION	EXPERT JUDGMENT WITH DATA SUPPO
Simplicity	High	Medium	Low
Accuracy	Moderate	High (if data is robust)	High (if experts are reliable)
Scalability	High	Medium	Low
Resource Requirements	Low	High	High
Adaptability	Low	Medium	High

Selected Approach

The best approach depends on the availability of data, resources, and the need for accuracy. However, **Machine Learning-Based Prediction** is the most robust and scalable option if sufficient data and computational resources are available. It balances accuracy and adaptability while leveraging advanced analytics to uncover hidden patterns.

Python Code for Selected Approach

Below is an example implementation of the Machine Learning-Based Prediction approach:

```
python
1 import pandas as pd
```



```

2 from sklearn.model_selection import train_test_split
3 from sklearn.ensemble import RandomForestRegressor
4 from sklearn.metrics import mean_absolute_error
5
6 # Load the dataset
7 file_path = 'sicor_municipal_contracts.csv'
8 data = pd.read_csv(file_path)
9
10 # Step 1: Preprocess the data
11 # Assume columns: 'institution_id', 'contract_value', 'date', etc.
12 data['date'] = pd.to_datetime(data['date']) # Convert date column to datetime
13 first_semester_data = data[data['date'].dt.month <= 6] # Filter for first semester
14
15 # Aggregate total investment per institution
16 institution_investment = first_semester_data.groupby('institution_id')['contract_value'].sum().reset_index()
17 institution_investment.columns = ['institution_id', 'total_investment']
18
19 # Add additional features (example)
20 institution_investment['num_contracts'] = first_semester_data.groupby('institution_id').size().reset_index()['size']
21 institution_investment['avg_contract_size'] = institution_investment['total_investment'] / institution_investment['num_contracts']
22
23 # Step 2: Prepare data for modeling
24 X = institution_investment[['num_contracts', 'avg_contract_size']] # Features
25 y = institution_investment['total_investment'] # Target variable
26
27 # Split data into training and testing sets
28 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
29
30 # Step 3: Train a Random Forest model
31 model = RandomForestRegressor(random_state=42)
32 model.fit(X_train, y_train)
33
34 # Step 4: Evaluate the model
35 y_pred = model.predict(X_test)
36 mae = mean_absolute_error(y_test, y_pred)
37 print(f"Mean Absolute Error: {mae}")
38
39 # Step 5: Predict future investments
40 future_predictions = model.predict(X) # Predict for all institutions
41 institution_investment['predicted_investment'] = future_predictions
42
43 # Step 6: Rank institutions and select top 10
44 top_10_institutions = institution_investment.sort_values(by='predicted_investment', ascending=False)
45 print(top_10_institutions[['institution_id', 'predicted_investment']])

```

Conclusion

The **Machine Learning-Based Prediction** approach is selected as the best method due to its ability to capture complex patterns and provide scalable predictions. The provided Python code demonstrates how to implement this approach using a Random Forest model. If additional data or expert insights are available, they can be incorporated to further enhance the predictions.

